

Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d -step measurement delay:

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= \mathbf{a}_{x_{det}}[\mathbf{k} + \mathbf{1}] + a_{x_{ram}}[k+1] \\ &= \underbrace{\left\{ \mathbf{a}_{x_{det}}[\mathbf{k}] + a_{x_{ram}}[k] \right\}}_{a_x[k]} + \Delta \mathbf{a}_x[\mathbf{k}] + \underbrace{\left\{ a_{x_{ram}}[k+1] - a_{x_{ram}}[k] \right\}}_{\Delta a_{ram}[k]} \end{aligned}$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k], \quad \Delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k] \text{ (desired-trajectory height increment).}$$

$$\text{Parameter estimator:} \quad \hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k]$$

$$\text{Parameter estimation error:} \quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$\text{Estimation error dynamics:} \quad e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k]$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \Delta a_{ram}[k-i] \right\}}_{\delta x_{\delta af}^d[k]} \\ &\quad + \delta x_D^d[k] + (1 - \lambda_c) n_x[k] \end{aligned}$$

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

State equation

($\delta x_D^d \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \left(\hat{a}_x[k] - \sum_{i=1}^d \Delta a_x[k-i] \right) \\ &= \left(a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + r_2[k] \right) - \hat{a}_x[k] + \sum_{i=1}^d \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{\Delta a_{ram}[k]}_{q_4[k]}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

Gain fluctuation variance $\text{Var}(\Delta a_{ram})$

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$.

$$a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}, \quad K_h = \frac{1}{c} \frac{dc}{d\bar{h}} \text{ (wall sensitivity)}, \quad a_{x_{ram}} = a'_x x_{ram}, \quad x_{ram} = -\delta x$$

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

Increment of a stationary signal ($\rho_1 = \text{Cov}(x_{ram}[k+1], x_{ram}[k]) / \text{Var}(x_{ram})$, lag-1 autocorr):

$$\text{Var}(\Delta x_{ram}) = 2 \text{Var}(x_{ram}) - 2 \text{Cov}(x_{ram}[k+1], x_{ram}[k]) = 2(1 - \rho_1) \text{Var}(x_{ram})$$

$$\rho_1 = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}} \Rightarrow 1 - \rho_1 = \frac{1}{(1 + \lambda_c) C_{\delta x}}$$

Gain increment $\Delta a_{ram} = a'_x \Delta x_{ram}$ ($C_{\delta x}$ and $(1 - \rho_1)$ cancel):

$$\text{Var}(\Delta a_{ram}) = a_x'^2 \text{Var}(\Delta x_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_{\delta h}^2$$

$$\text{Var}(\Delta a_{ram}) = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2$$

Impl.: K_h , $\sigma_{\delta h}^2 = 4k_B T a_\perp$ evaluated at $\bar{h}_{\text{meas}}$ (measured, p_m); gain $a_x \rightarrow \hat{a}_x$ (estimate).

Correction — the R_{22} delay term telescopes

$$\Delta a_{ram}[k-1] + \Delta a_{ram}[k-2] = a_{ram}[k] - a_{ram}[k-2] \Rightarrow \text{Var}\left(\sum_{i=1}^2 \Delta a_{ram}[k-i]\right) = 2(1 - \rho_2) \text{Var}(a_{ram})$$

$$R_{22}^{\text{delay}} = \frac{1 - \rho_2}{2(1 - \rho_1)} \sum_{i=1}^d \text{Var}(\Delta a_{ram}[k-i]), \quad \frac{1 - \rho_2}{2(1 - \rho_1)} = 1.105 \text{ } (\lambda_c=0.7)$$

$$\rho_1 = 0.8515, \quad \rho_2 = 0.6718 \quad (\rho_\tau = \text{lag-}\tau \text{ autocorr of } \delta x; \text{ independent sum} \Rightarrow -9.5\%)$$

Impl.: offline constant `r22_delay_sum_factor` (`build_eq17_6state_constants`); $d = 1 \rightarrow 1$ (single term, exact).

Correction — $a_{x_{ram}}$ is first-order autoregressive (AR(1)), not a random walk

True-signal dynamics ($x_{ram} = -\delta x$; $\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k]$):

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = \lambda_c a_{x_{ram}}[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]$$

$$\Delta a_{ram}[k] = (\lambda_c - 1) a_{x_{ram}}[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]$$

True gain state ($a_x = a_{x_{det}} + a_{x_{ram}}$, $a_{x_{det}}[k+1] = a_{x_{det}}[k] + \Delta a_x[k]$, $e_{a_x} = a_x - \hat{a}_x$):

$$\begin{aligned} a_x[k+1] &= (a_{x_{det}}[k] + \Delta a_x[k]) + \lambda_c (a_x[k] - a_{x_{det}}[k]) + a'_x F_{dx}[k] (a_x[k] - \hat{a}_x[k]) + a'_x \varepsilon[k] \\ &= \underbrace{(\lambda_c + a'_x F_{dx}[k]) a_x[k]}_{\mathbf{F}_e(4,4)} + \underbrace{(1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] - a'_x F_{dx}[k] \hat{a}_x[k]}_{\text{known input}} + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e(4,4) = \lambda_c + a'_x F_{dx}[k]}$$

Estimator ($(\lambda_c + a'_x F_{dx}) \hat{a}_x - a'_x F_{dx} \hat{a}_x = \lambda_c \hat{a}_x$: the feedthrough vanishes at the estimate):

$$\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + \ell_{4\cdot}[k] e_y[k]$$

$$e_{a_x}[k+1] = a_x[k+1] - \hat{a}_x[k+1] = (\lambda_c + a'_x F_{dx}[k]) e_{a_x}[k] + a'_x \varepsilon[k] - \ell_{4\cdot}[k] e_y[k]$$

Process noise (white driver $w_4 = a'_x \varepsilon$; equivalent-white $\text{Var}_{\text{eff}}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$, stationary-variance matching — the per-step $\gamma_\varepsilon(0) = (1 + 2(1 - \lambda_c)^2) \sigma_{\delta h}^2$ differs by 1.712; ε is MA(d), δx is not AR(1)):

$$q_4 = \text{Var}(w_4) = a_x'^2 \text{Var}(\varepsilon) = a_x'^2 (1 - \lambda_c^2) \text{Var}(\delta x) = (1 - \lambda_c^2) C_{\delta x} a_x'^2 \sigma_{\delta h}^2$$

$$\boxed{q_4 = a_x'^2 \text{Var}(\varepsilon) = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2}$$

Impl.: predict and q_4 use $\hat{a}_x$ with a'_x at $\bar{h}_{\text{meas}}(p_m)$; reversion target $a_{x_{det}}$ from p_d ; production $\mathbf{F}_e(4,4) = \lambda_c$ (drops $a'_x F_{dx}$).

Taylor gain — the full first-order model (anchor-free)

Gain by Taylor expansion about the desired trajectory ($x = x_d - \delta x$):

$$a_x[k+1] = a_x(x[k+1]) = \underbrace{a_x(x_d[k+1])}_{a_{x_{det}}[k+1]} - a'_x[k+1] \delta x[k+1], \quad a_{x_{det}}[k+1] = a_{x_{det}}[k] + a'_x[k] \Delta h_d[k], \quad a'_x[k+1] \approx$$

True gain state ($a_{x_{det}}[k] = a_x[k] + a'_x \delta x[k]$, then the boxed δx recursion):

$$\boxed{a_x[k+1] = a_x[k] + a'_x \Delta h_d[k] + (1 - \lambda_c) a'_x \delta x[k] + a'_x F_{dx}[k] e_{a_x}[k] + a'_x \varepsilon[k]}$$

Estimator ($E\{e_{a_x}\} = 0$, $E\{\varepsilon\} = 0$; $\delta x[k] \rightarrow \delta \hat{x}_3[k]$):

$$\boxed{\hat{a}_x[k+1] = \hat{a}_x[k] + a'_x \Delta h_d[k] + (1 - \lambda_c) a'_x \delta \hat{x}_3[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]}$$

Error dynamics ($e_{\delta x_3} = \delta x_3 - \delta \hat{x}_3$):

$$\boxed{e_{a_x}[k+1] = (1 + a'_x F_{dx}[k]) e_{a_x}[k] + (1 - \lambda_c) a'_x e_{\delta x_3}[k] + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} - \ell_{41}[k] e_{y_1}[k] - \ell_{42}[k] e_{y_2}[k]}$$

F_e and process noise ($q_3 = -\varepsilon$, $q_4 = a'_x \varepsilon$: one sample $\Rightarrow$ rank-1):

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & (1 - \lambda_c) a'_x & 1 + a'_x F_{dx}[k] \end{bmatrix}, \quad \text{eig}(\mathbf{F}_e) = \{0, 0, \lambda_c + a'_x F_{dx}, 1\}}$$

$$\boxed{Q_{33} = \sigma_\varepsilon^2, \quad Q_{34} = Q_{43} = -a'_x \sigma_\varepsilon^2, \quad Q_{44} = a'^2_x \sigma_\varepsilon^2, \quad \sigma_\varepsilon^2 := \gamma_\varepsilon(0) = (1 + 2(1 - \lambda_c)^2) \sigma_{\delta h}^2}$$

$$P_{44} = a'^2_x P_{33}, \quad P_{34} = -a'_x P_{33} \quad (\text{exact filter invariant, } \forall R_{22})$$

$$\frac{\gamma_\varepsilon(0)}{1 - \lambda_c^2} = 2.31 \sigma_{\delta h}^2 \quad \text{vs} \quad \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2 = 3.96 \sigma_{\delta h}^2 \quad (\varepsilon \text{ colored: white-}\varepsilon \text{ model under-represents } \text{Var}(a_{ram}))$$

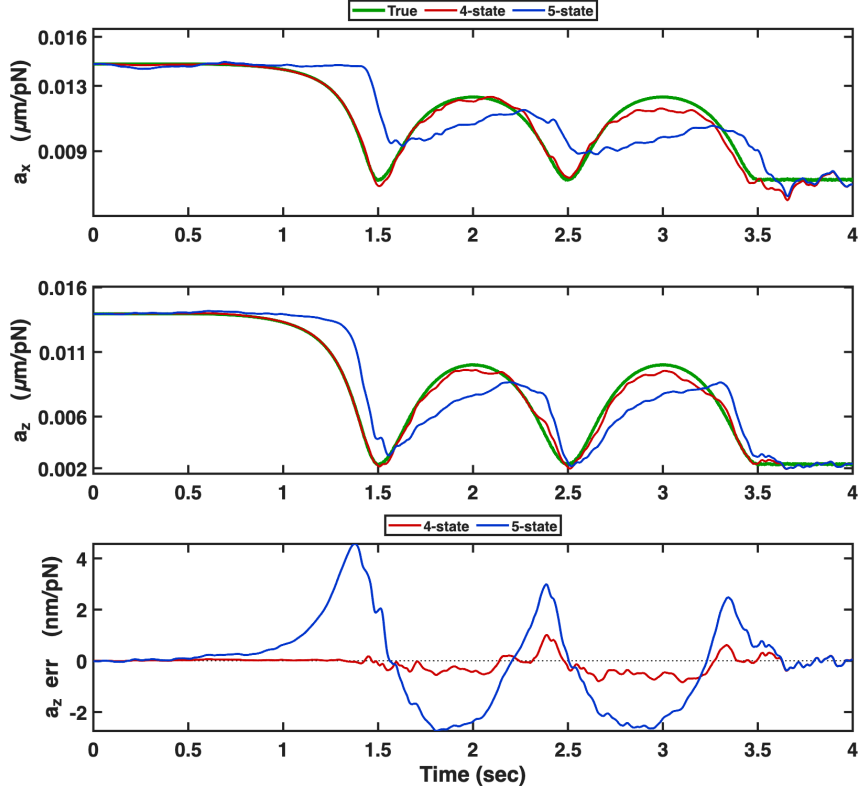
Feedforward (implemented forms; raw $a' \Delta h$ accumulates $\sim 3.6\%$ Euler drift over the descent):

$$\text{known: } \Delta \hat{a} = a_{x_{det}}[k] - a_{x_{det}}[k-1]; \quad \text{ahat: } \Delta \hat{a} = \hat{a}_x \left(\frac{a_{x_{det}}[k]}{a_{x_{det}}[k-1]} - 1 \right); \quad \text{diff: } \Delta \hat{a} = \hat{a}'_x \Delta h_d[k-1]$$

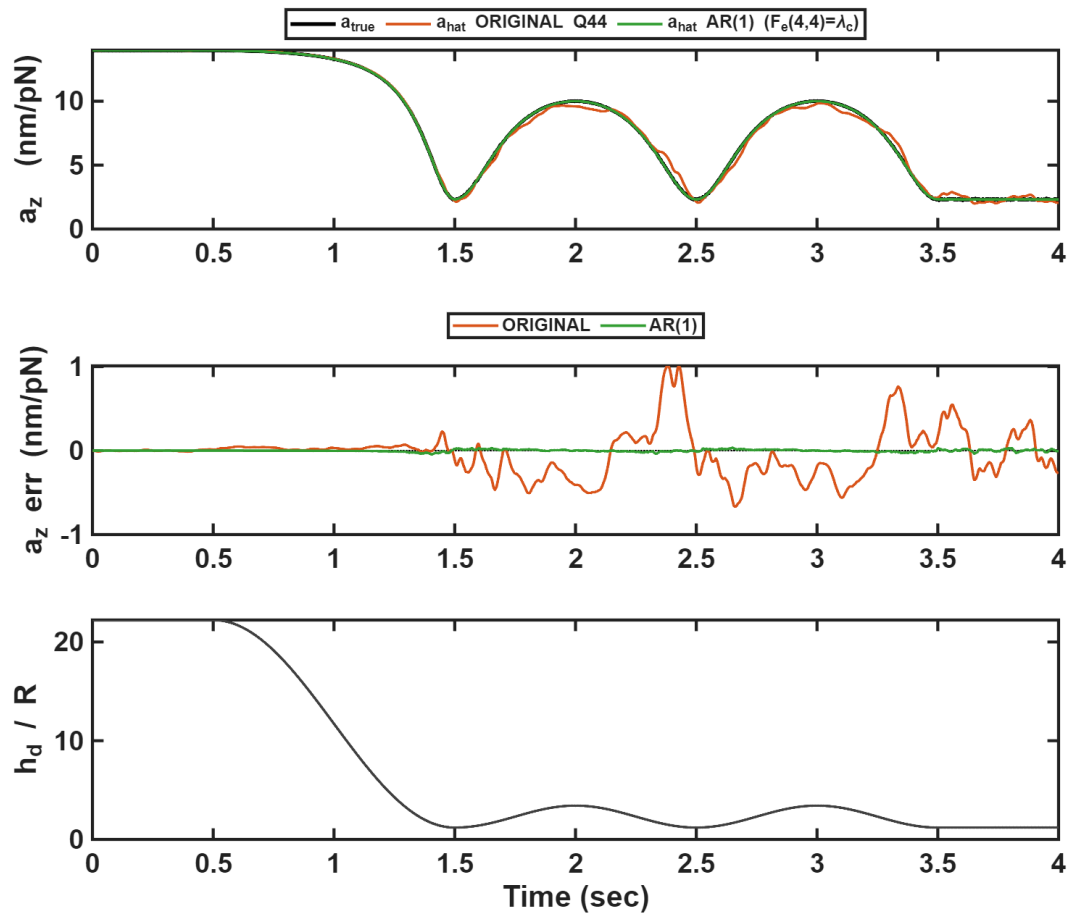
Impl.: knobs `use_taylor_gain` + `aprime_source` (known | ahat | diff = the self-difference box below), default off. z axis: full row 4. x, y : the gain argument is the wall-normal height, so z 's $\delta \hat{x}_3$ enters their predict as a known input; row 4 = $[0 \ 0 \ 0 \ 1]$, $Q_{34} = 0$, $Q_{44} = a'^2_i \sigma_{\varepsilon, z}^2$.

Verification (1 Hz, near wall)

Per-axis EKF, seed ensemble; near-wall 1 Hz oscillation ($h_{init} = 50 \mu\text{m}$, trough $\bar{h} = 1.2$).



a_x, a_z : true gain and the 4-state / 5-state estimates; bottom tile, the a_z estimation error $\hat{a}_x - a_x$.



a_z : original random-walk Q_{44} ($F_e(4,4) = 1$) vs $AR(1)$ reverting gain ($F_e(4,4) = \lambda_c$); middle tile, estimation error; bottom, $\bar{h}$. Near-wall $|\hat{a}_z - a_z|$: 9.4% $\rightarrow$ 0.4%.

Model-free a' via self-difference — original (RW) vs AR(1)

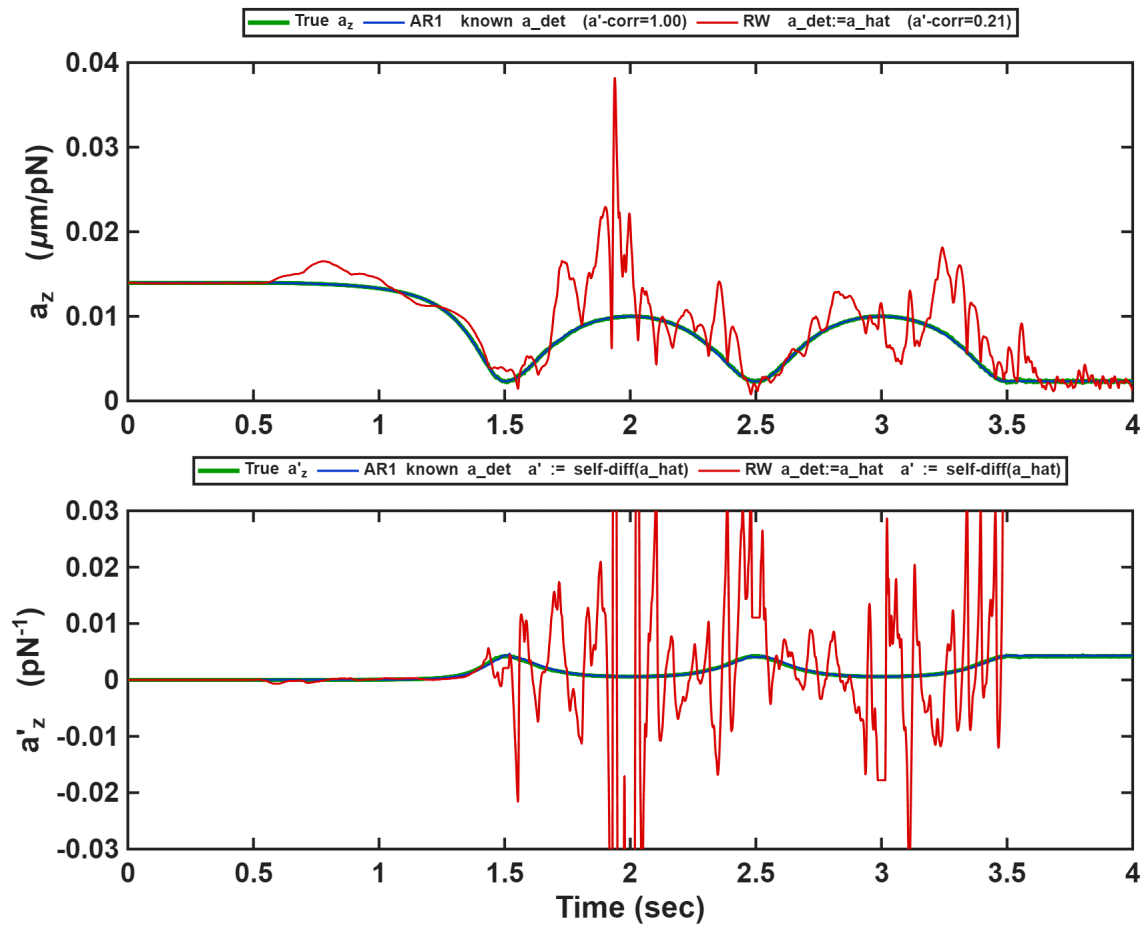
So far a'_x has come from the known wall model. If the wall is unknown, we can read a'_x straight off the gain estimate instead: take the change in $\hat{a}_x$ and divide it by the (known) change in desired height. But that raw ratio is very jumpy — $\hat{a}_x$ already carries estimation noise, and taking a difference makes the noise worse. So we do two simple things: (i) smooth the raw slope with a running average (an EWMA with weight β), and (ii) stop updating whenever the height is barely moving (a gate γ , so we never divide by almost zero):

$$\hat{a}'_x[k] = \begin{cases} (1 - \beta) \hat{a}'_x[k-1] + \beta \frac{\hat{a}_x[k] - \hat{a}_x[k-1]}{\Delta h_d[k-1]}, & |\Delta h_d[k-1]| > \gamma \\ \hat{a}'_x[k-1], & |\Delta h_d[k-1]| \leq \gamma \quad (\text{gate closed: keep the last value}) \end{cases}$$

$\beta = 0.05$ (running-average weight: smaller = smoother but slower) $\gamma = 10^{-3} \mu\text{m}$ (freeze $\hat{a}'$ once $|\Delta h_d|$ drops below this)

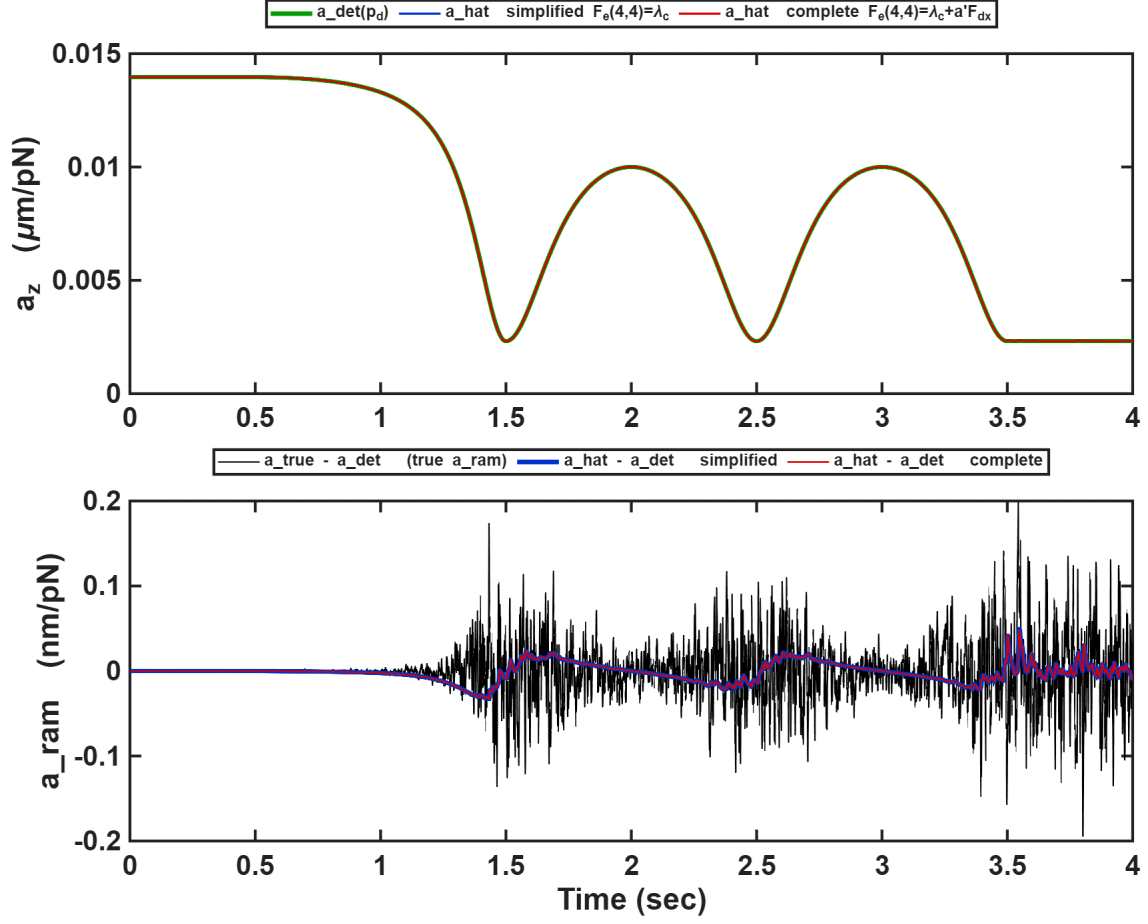
This one self-difference supplies a' everywhere in both runs; the runs differ *only* in how the gain *level* $a_{x_{det}}$ is anchored:

- **original (RW):** $a_{x_{det}} := \hat{a}_x$ (no outside anchor — the estimate leans on itself), $\mathbf{F}_e(4, 4) = 1$, $Q_{44} = \frac{2}{1 + \lambda_c} \hat{a}_x'^2 \sigma_{\delta h}^2$ — *fully* model-free.
- **AR(1):** the level $a_{x_{det}} = a_{\text{nom}}/c(\bar{h}_d)$ is still taken from the wall, $\mathbf{F}_e(4, 4) = \lambda_c$, $Q_{44} = (3 - 2\lambda_c^2) \hat{a}_x'^2 \sigma_{\delta h}^2$ — a' is model-free, but the level still needs the wall.



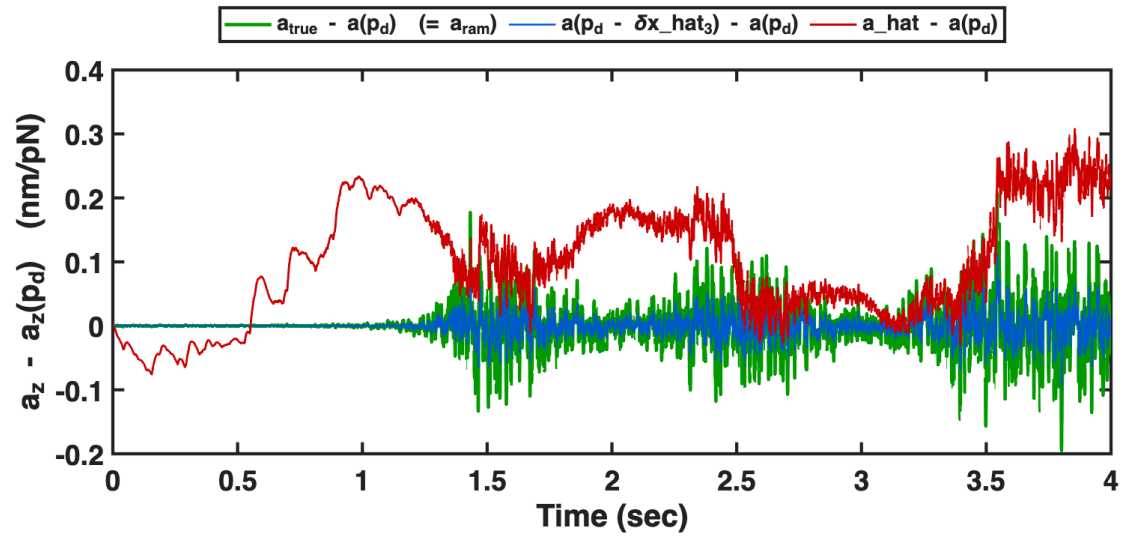
Same self-difference $\hat{a}'_x$ in both runs. Top: gain a_z . Bottom: slope a'_z . With a known level anchor (AR(1)) the estimate follows the truth; without one (RW) it is buried in noise.

Verification — $F_e(4,4) = \lambda_c$ (simplified) vs $\lambda_c + a'_x F_{dx}$ (complete), known $c(\bar{h})$, same seed

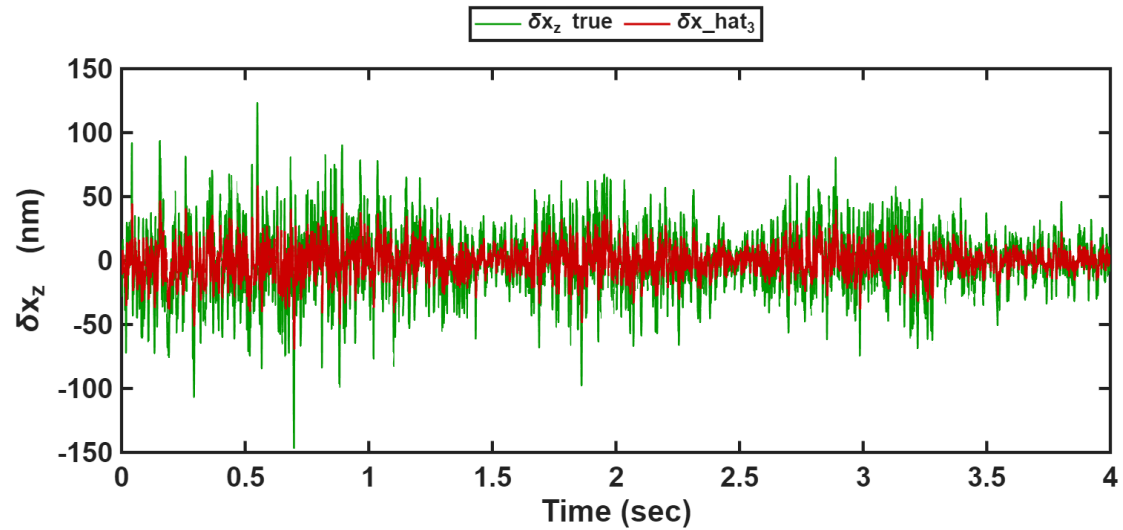


Top: $a_z - a_{x_{\text{det}}}(p_d)$ and the two $\hat{a}_x$. Bottom: $a_x - a_{x_{\text{det}}}$ (true, black) and $\hat{a}_x - a_{x_{\text{det}}}$ (both versions). The two versions coincide: $\max |\hat{a}_x^{\text{complete}} - \hat{a}_x^{\text{simplified}}| / a_{x_{\text{det}}} = 0.085\%$; $a_{x_{\text{det}}}(p_d) - \hat{a}_x = -0.03\% \pm 0.30\%$ ($\bar{h}_d < 2$: $\pm 0.41\%$).

Verification — taylor gain (z axis, 1 Hz osc, seed 1)



Known- a' run: $a_{x,\text{true}} - a_x(p_d) (= a_{x_{\text{ram}}})$, $a_x(p_d - \delta \hat{x}_3) - a_x(p_d) (\approx -a'_x \delta \hat{x}_3)$, and $\hat{a}_x - a_x(p_d)$.



Same run: $\delta \hat{x}_3$ vs true δx_z .

Verification — e_{a_x} injection into the motion error (two runs, same noise)

Two closed-loop runs driven by the same (f_T, n_x) sequences; the superscript is the gain used by the control law — $(\hat{a})$: the estimate $\hat{a}_x$; (a) : the true a_x (so e_{a_x} does not enter the control):

$$\delta x^{(\hat{a})}[k+1] = \lambda_c \delta x^{(\hat{a})}[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k]$$

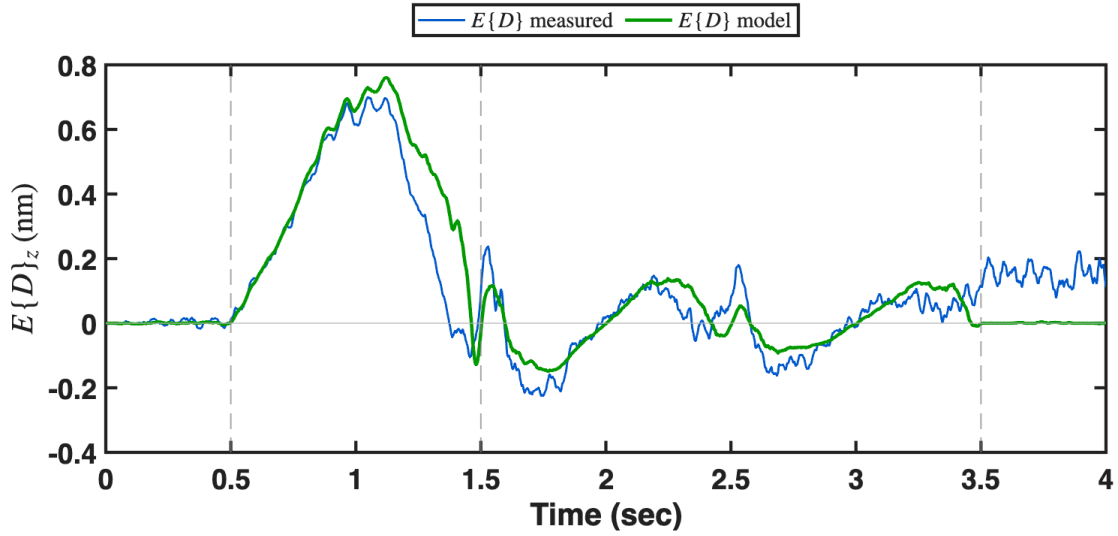
$$\delta x^{(a)}[k+1] = \lambda_c \delta x^{(a)}[k] - \varepsilon[k]$$

Difference of the two runs (the common ε cancels):

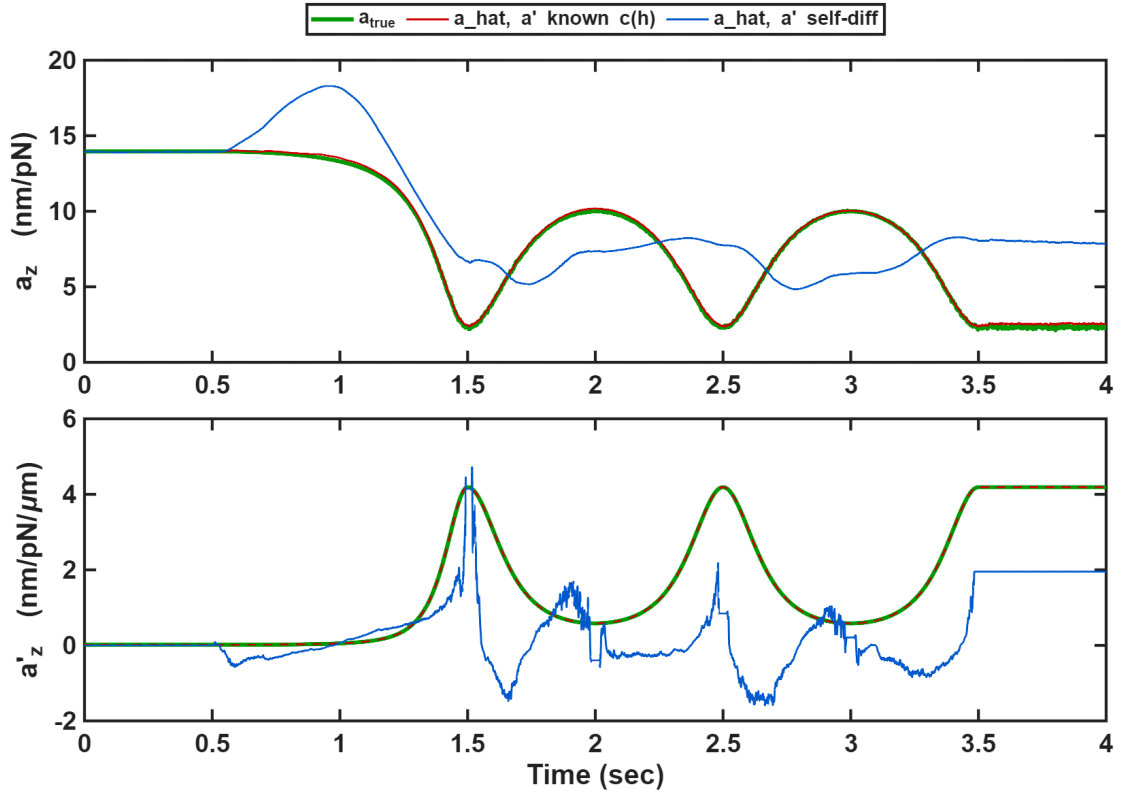
$$D := \delta x^{(\hat{a})} - \delta x^{(a)}, \quad D[k+1] = \lambda_c D[k] - F_{dx}[k] e_{a_x}[k]$$

Mean over seeds ($E\{F_{dx} e_{a_x}\} \approx E\{F_{dx}\} E\{e_{a_x}\}$, Cov term measured 0.03 nm):

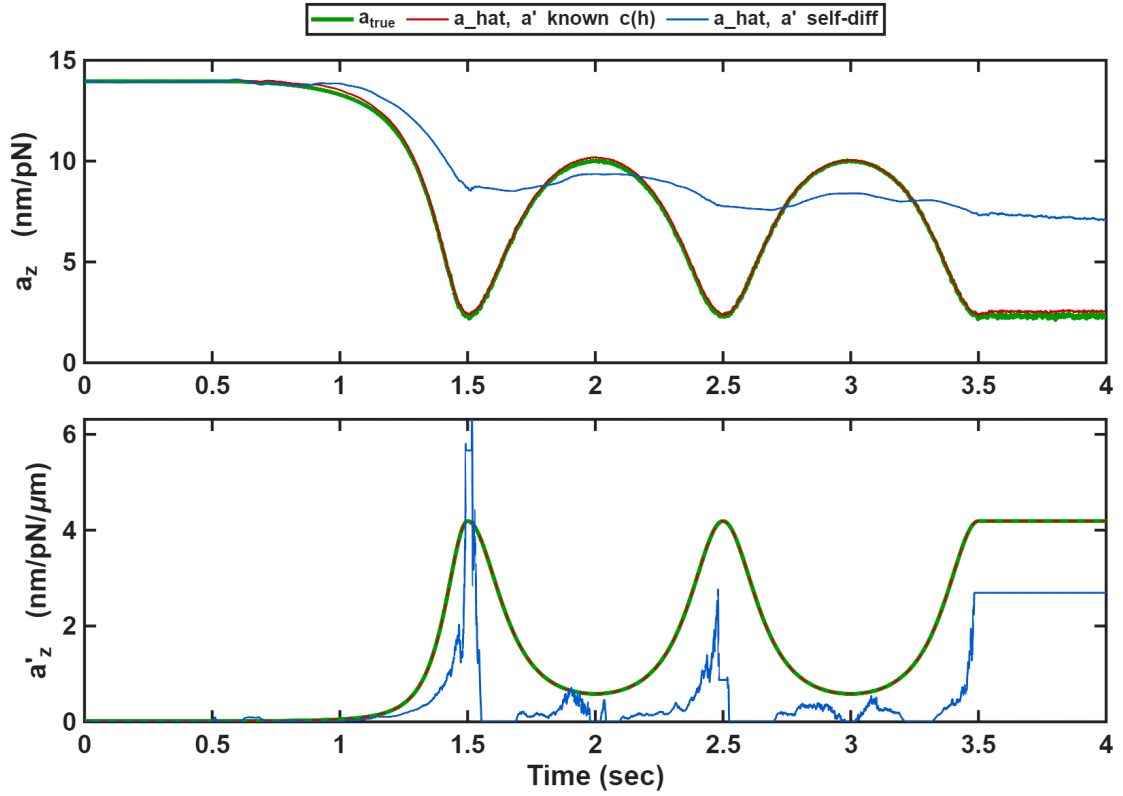
$$E\{D\}[k+1] = \lambda_c E\{D\}[k] - E\{F_{dx}\}[k] E\{e_{a_x}\}[k]$$



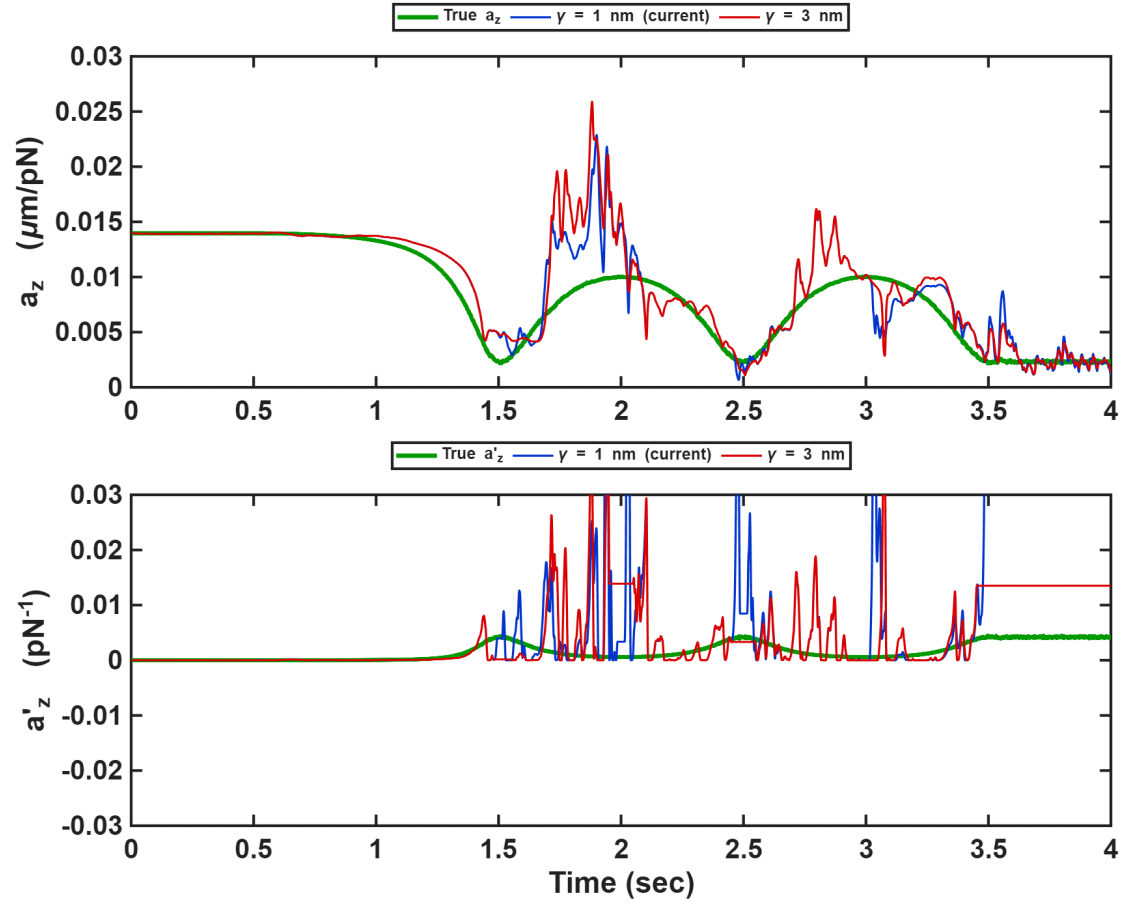
z axis, 100 seeds, 26 ms moving average. Measured $E\{D\}$ vs the recursion above driven by the measured $E\{e_{a_x}\}$, $E\{F_{dx}\}$: slope 0.850, corr 0.905. Hold: $F_{dx} \approx 0 \Rightarrow E\{D\} = 0$. The 15% amplitude gap is second order $(\delta x_{\delta a_f}^d)$.



$\hat{a}_x$ (top) and a'_x (bottom): known a' vs self-diff a' (no clamp).



Same seed with the post-EWMA clamp $\hat{a}'_x \leftarrow \max(\hat{a}'_x, 0)$. Six-arm study (20 seeds, osc window): taylor-known $-0.69\% / 3.08\%$ vs RW $-2.5\% / 16.5\%$ vs AR(1) $+0.02\% / 0.96\%$ (bias/std); self-diff $+33\% / 90\%$.



Self-diff gate sweep: $\gamma = 1$ nm (current) vs $\gamma = 3$ nm — $\hat{a}_x$ (top) and $\hat{a}'_x$ (bottom); the flat segments are the gate holding the last slope.

e_{a_x} : the combination $v := -(e_{a_x} + a'_x e_{\delta x_3})$

$$\begin{aligned} e_{\delta x_3}[k+1] &= \lambda_c e_{\delta x_3} - F_{dx} e_{a_x} - \varepsilon - \ell_{31} e_{y_1} - \ell_{32} e_{y_2} \\ e_{a_x}[k+1] &= (1 + a'_x F_{dx}) e_{a_x} + (1 - \lambda_c) a'_x e_{\delta x_3} + a'_x \varepsilon - \ell_{41} e_{y_1} - \ell_{42} e_{y_2} \end{aligned}$$

row 2 + $a'_x \times$ row 1:

$$(e_{a_x} + a'_x e_{\delta x_3})[k+1] = (e_{a_x} + a'_x e_{\delta x_3})[k] - (\ell_{41} + a'_x \ell_{31}) e_{y_1} - (\ell_{42} + a'_x \ell_{32}) e_{y_2}$$

$$v := -(e_{a_x} + a'_x e_{\delta x_3}) = \hat{a}_x - (a_{x_{det}} - a'_x \delta \hat{x}_3), \quad e_{a_x} = -a'_x e_{\delta x_3} - v$$

$$\boxed{v[k+1] = v[k] + g_1 e_{y_1}[k] + g_2 e_{y_2}[k] + (a'_x[k+1] - a'_x[k]) \delta \hat{x}_3[k], \quad g_j := \ell_{4j} + a'_x \ell_{3j}}$$

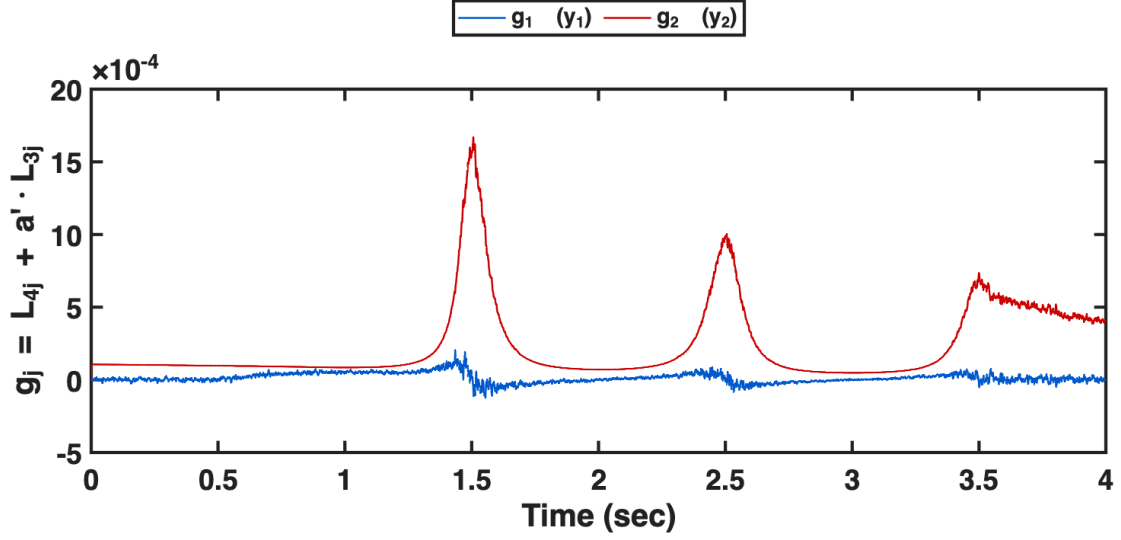
$$\ell_{31} = \frac{P_{31}}{S_1}, \quad \ell_{41} = \frac{P_{41}}{S_1}, \quad \ell_{32} = \frac{P_{34}}{S_2}, \quad \ell_{42} = \frac{P_{44}}{S_2} \implies g_1 = \frac{P_{41} + a'_x P_{31}}{S_1}, \quad g_2 = \frac{P_{44} + a'_x P_{34}}{S_2}$$

$$g_1 = 0 \iff P_{41} = -a'_x P_{31}, \quad g_2 = 0 \iff P_{44} = -a'_x P_{34}$$

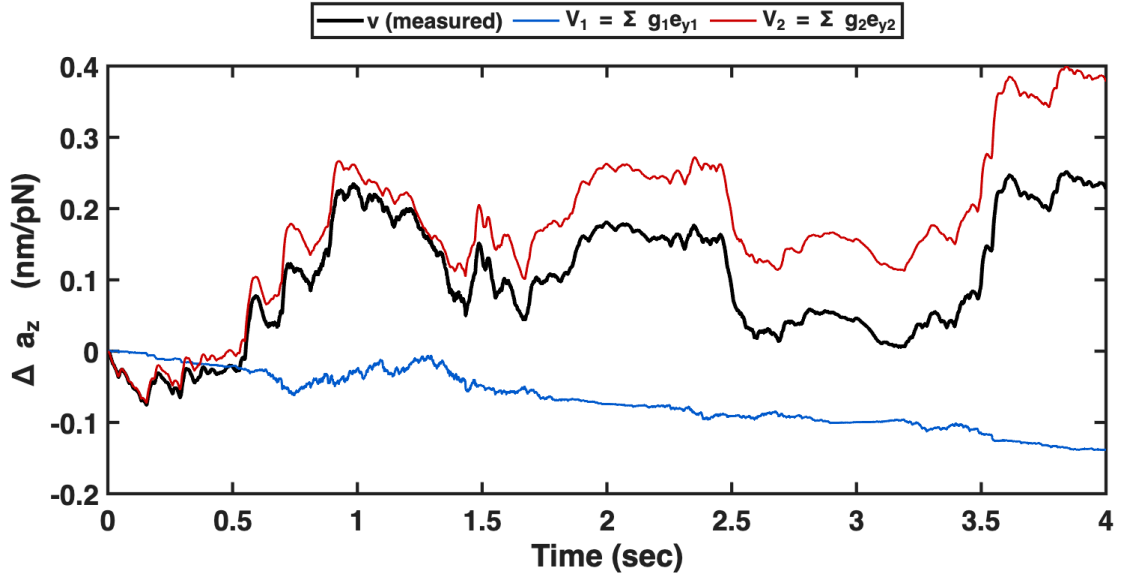
$$v[k] = \underbrace{\sum g_1 e_{y_1}}_{V_1} + \underbrace{\sum g_2 e_{y_2}}_{V_2} + \underbrace{\sum \Delta a'_x \delta \hat{x}_3}_{C_3}$$

Impl.: diag adds innovation_y1, K_kf_a_y1, K_kf_dx_y2; budget harness verify_taylor_v_budget.m.

Verification — v budget (z axis, 1 Hz osc, seed 1, oracle- a' arm)



$g_1 \approx 0$: the y_1 correction keeps $\ell_{41} \approx -a'_x \ell_{31}$, so it barely moves v ; g_2 peaks 1.7×10^{-3} at the troughs and stays $\sim 5 \times 10^{-4}$ after 3.5 s.



Measured v vs the two running sums ($C_3 - 0.012$ not shown; $V_1 + V_2 + C_3 - v = 7 \times 10^{-14}$). At $t=4$ s: $V_2 + 0.378$, $V_1 - 0.139$, $v + 0.227$ nm/pN — v comes from y_2 (a_{xm}); y_1 cancels about $1/3$.

Colored y_2 — measurement-noise dynamics

$$y_2[k] = a_{x_m}[k] = a_x[k] + v[k], \quad \text{KF: } \mathbb{E}\{v[k]v[j]\} = R_{22} \delta_{kj}$$

$$\hat{\sigma}_{\delta x_r}^2[k+1] = (1 - a_{cov}) \hat{\sigma}_{\delta x_r}^2[k] + a_{cov} \delta x_r^2[k+1], \quad a_{x_m}[k] = \frac{\hat{\sigma}_{\delta x_r}^2[k] - C_n \sigma_{n_x}^2}{C_{dpmr} 4k_B T}$$

$$\tilde{v}[k+1] = \alpha \tilde{v}[k] + a_{cov} \tilde{s}[k+1]; \quad \alpha = 1 - a_{cov}, \quad \tilde{s}[k] = \delta x_r^2[k] - \mathbb{E}\{\delta x_r^2\}$$

$$\rho_s(\tau) = \rho_{\delta x_r}^2(\tau) \quad (\delta x_r \sim \mathcal{N}, \text{ Isserlis})$$

ACF of v

$$\gamma_v(\tau) = a_{cov}^2 \sum_{i,j \geq 0} \alpha^{i+j} \gamma_s(\tau + i - j) = \frac{a_{cov}^2}{1 - \alpha^2} \sum_{m=-\infty}^{\infty} \alpha^{|m|} \gamma_s(|\tau + m|)$$

$$\rho_v(\tau) = \frac{\sum_m \alpha^{|m|} \rho_s(|\tau + m|)}{\sum_m \alpha^{|m|} \rho_s(|m|)}, \quad \varphi := \rho_v(1) = 0.9864 \quad (\lambda_c = 0.7, \ a_{pd} = a_{cov} = 0.05)$$

$$\rho_v(\tau+1)/\rho_v(\tau) \xrightarrow{\tau \rightarrow \infty} \alpha = 0.95$$

τ	1	2	5	10	20	50
$\rho_v(\tau)$ analytic	0.9840	0.9512	0.8271	0.6513	0.4040	0.0905
measured (A2, 20 seeds)	0.9854	0.9526	0.8260	0.6472	0.4013	0.1008

$$\frac{(1 - \varphi^2) \gamma_v(0)}{\gamma_v(0)} = 1 - \varphi^2 \approx \frac{1}{37}$$

AR(1) noise state

$$v[k+1] = \varphi v[k] + w_v[k], \quad x_e = [\delta x_1, \delta x_2, \delta x_3, a_x, v]^\top$$

$$\mathbf{F}_e \rightarrow \text{blkdiag}(\mathbf{F}_e^{4 \times 4}, \varphi), \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$Q_{55}[k] = (1 - \varphi^2)(1 - r_f) R_{2,\text{int}}[k], \quad R_{22}[k] = r_f R_{2,\text{int}}[k] + R_{2,\text{delay}}[k], \quad r_f = 0.05$$

$$\text{Var}(v) + R_{22} = R_{2,\text{int}}[k] + R_{2,\text{delay}}[k]$$

Init hold (first d samples)

$$k \leq d: \quad \delta x_m[k] \equiv 0 \quad (\text{buffer init}) \Rightarrow \hat{\sigma}_{\delta x_r}^2[k] := \hat{\sigma}_{\delta x_r}^2[0], \quad \overline{\delta x}_m[k] := \overline{\delta x}_m[0]$$

F_e endogeneity — expectation linearization (f_{det})

$$f_{dx}[k] = f_{det}[k] + f_{ram}[k]$$

Deterministic mirror (control law at $\delta x_m := 0$):

$$f_{det}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] - (1 - \lambda_c) \sum_{i=1}^d \hat{a}_x[k-i] f_{det}[k-i] \right\}$$

Remainder (linear in the noise history):

$$f_{ram}[k] = \hat{a}_x^{-1}[k] (1 - \lambda_c) \left\{ \delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{ram}[k-i] \right\}$$

$$F_{det}[k] = f_{det}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{det}[k-i], \quad F_{ram}[k] = f_{ram}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{ram}[k-i],$$

$$F_{dx} = F_{det} + F_{ram}$$

Violation

$$\delta x[k+1] = \lambda_c \delta x[k] - F_{dx}[k] e_{a_x}[k] - \varepsilon[k]$$

$F_{ram}[k] = F_{ram}(\{n_x, f_T\}_{<k})$: same noises driving $\varepsilon[k]$ and the innovation $\nu[k]$.

$$\text{Cov}(F_{ram}[k], \varepsilon[k]) \neq 0 \Rightarrow K[k] = K(F_{det} + F_{ram}) \not\perp \nu[k]$$

$$\Rightarrow E\{K \nu\} = \bar{K} E\{\nu\} + \text{Cov}(K, \nu) \neq 0$$

$\text{Cov}(K, \nu)$ term : per-step systematic drift on $\hat{a}_x$ (rectification via filter geometry, not via R)

Fix — expectation linearization

F_e built from $F_{det} = E\{F_{dx} \mid \text{trajectory, } \hat{a}_x \text{ history}\}$; the error dynamics rewrites exactly as

$$\delta x[k+1] = \lambda_c \delta x[k] - F_{det}[k] e_{a_x}[k] - \tilde{\varepsilon}[k],$$

$$\tilde{\varepsilon}[k] := \varepsilon[k] + F_{ram}[k] e_{a_x}[k]$$

$F_{det}[k]$ known (trajectory, $\hat{a}_x$ history) $\Rightarrow$ KF independence restored; $F_{ram}[k] e_{a_x}[k] \rightarrow$ noise

Residual magnitude ($\rightarrow Q$, negligible)

$$\text{Var}(F_{ram}[k] e_{a_x}[k]) \approx \text{Var}(F_{ram}[k]) E\{e_{a_x}^2[k]\} \quad (F_{ram} \perp e_{a_x} \text{ to first order})$$

$$\frac{\text{Var}(F_{ram} e_{a_x})}{\text{Var}(\varepsilon)} < 10^{-3} \quad (\lambda_c=0.7, a_{pd}=a_{cov}=0.05, \text{ hold, } z, e_{a_x} \sim 0.03 a_x)$$

Verification

(colored- y_2 ‘ar1’ model, 20 seeds, z axis, osc window)

arm	bias (%) before → after	std (%) before → after	div before → after
A2 (a' known)	$-3.49 \rightarrow -0.37$	$3.04 \rightarrow 2.53$	$0/21 \rightarrow 0/21$
A3 ($a' = \hat{a}_x K_h$)	$-4.52 \rightarrow -1.59$	$2.42 \rightarrow 2.26$	$0/21 \rightarrow 0/21$
A4 (a' self-diff, surv.)	$+6.4 \rightarrow +14.1$	$66 \rightarrow 71$	$10/21 \rightarrow 1/21$
A1 (RW baseline)	$-1.73 \rightarrow +0.30$	$16.7 \rightarrow 6.13$	$0/21 \rightarrow 0/21$
A0 (production AR(1))	$+0.02 \rightarrow +0.03$	$0.96 \rightarrow 0.96$	$0/21 \rightarrow 0/21$

white (deaf) control: A2 $-0.43 \rightarrow +0.29$ (drift $\propto$ gain $\Rightarrow$ no effect)

$P_{44}/\text{Var}(e_{a_x}) = 0.415 \rightarrow 0.396$ (honesty unchanged $\Rightarrow$ independent problem)